A sparse B_4 compiler for the ten-integer Con(ZFC) equation

Research supplement for ProveIt

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Abstract

The pinned ten-integer construction uses base-five digit positions to separate quartic monomials. That separation is substantially stronger than necessary. Replacing it by an explicitly certified four-sum-distinct set preserves existence of integer solutions through the source interface. For the same fixed 31-variable quartic and the same 79 fixed circuit leaves, the resulting ten-integer equation has exact total degree 69 554 088 184 and a division-free evaluator of 1,407 binary operations, compared with degree 589 042 901 992 797 851 562 527 064 and 1,442 operations in the pinned construction. This is a source-encoding improvement, not a new ten-unknowns theorem, not a reduction in coefficient bit size, and not a claim of global optimality. The Lucas bridge and calibrated compiler endpoint remain explicit external dependencies.

1 Fixed input and exact result

The baseline is the directory `31-sun10/signed10` at ProveIt commit `22e75dd6b87b1b199bb08c778d7cf075ae5aae08`. All four native ports are kept together, with binding `wang943-mixed-121x8801`. In particular, the native port called `beta` must not be confused with the auxiliary Sun scale β_S below.

Let $r_1, \dots, r_{22}$ be the pinned quadratic residuals in 31 positive variables and set

$$P(n_1, \dots, n_{31}) = \sum_{j=1}^{22} r_j (n_1 + 1, \dots, n_{31} + 1)^2.$$

This definition, its fixed ports, and all its coefficients are retained in `sources/data`. The local verifier reconstructs the coefficient recipes from this definition. The inherited calibration is

$$\mathrm{ZFC} \vdash \left[(\exists n \in \mathbb{N}^{31})\ P(n) = 0 \right] \iff \neg\, \mathrm{Con}(\mathrm{ZFC}). \tag{1}$$

Equation (1) is an imported mathematical endpoint, not a claim that the Python checks establish a formal compiler-correctness proof.

Write $P = \sum_{|\alpha| \le 4} p_\alpha n^\alpha$ and

$$c_\alpha = (4 - |\alpha|)! \prod_{j=1}^{31} \alpha_j!\, p_\alpha, \qquad H_0 = \max_\alpha |p_\alpha|.$$

There are 402 nonzero weighted coefficient records. The data includes an unused leading dummy coordinate $\alpha_0 = 0$; it is not a coded variable. The fifth residual is 1 at the all-one assignment, so $P(0) > 0$ and $H_0 \ge 1$. The audited bound $|c_\alpha| + 1 < 2^{4269855}$ permits

$$\beta_S = 2^{4269882} > 48 \cdot 33^4 H_0.$$

It is an even power of two and is identical to the baseline scale.

2 Replacing exponential positions by a finite certificate

Use the following 31 positive weights, in the order of the parent variables:

51609	55430	68596	78030	89463
90233	110498	110508	128615	136498
211819	241519	305379	312706	314312
369115	417382	429683	435112	450230
462314	469608	490199	501539	507151
524944	550792	559943	573626	607799
610895				

Put $t = 610895$ and $L_c = 4t = 2443580$.

Lemma 1 (The finite separation certificate)**.** *The map $\alpha \mapsto e(\alpha) = \sum_j \alpha_j w_j$ is injective on $\{\alpha \in \mathbb{N}^{31} : |\alpha| \le 4\}$.*

Proof. Add the position zero and pad each multiset with zeros to length four. The 52,360 nondecreasing four-tuples of indices from these 32 positions have distinct integer sums. This is verified exhaustively by `verify_sun_b4.py`, using exact integers and no random sampling. Thus it is a finite, fully reproducible certificate of the asserted lemma.

For provenance, the candidate set comes from the Bose–Chowla construction in the field with binary polynomial $x^{20} + x^3 + 1$. The generator has a full orbit of $2^{20} - 1$ nonzero elements; the subfield has 32 elements. Taking logs of $\theta + a$, first subtracting 535448, multiplying by the unit 208418 modulo $2^{20} - 1$, and then subtracting 836756 gives the displayed positions. Equal four-term log sums would give two monic quartics in θ with the same value; their difference has degree at most three over the 32-element subfield, so it is zero and the multisets agree. The construction is supplementary: the direct finite check alone proves the particular integer claim used here. □

3 A generalized sparse source lemma

All sums and exponents in what follows have fixed integer bounds. Define

$$
\begin{aligned}
b &= 1 + 3f, & \mathcal{B} &= \beta_S b^4, & \mathscr{C} &= b\mathcal{B}^t,\\
\ell &= \sum_{j=1}^{31} \mathcal{B}^{w_j}, & D_c &= \sum_{|\alpha|\le 4} c_\alpha \mathcal{B}^{L_c - e(\alpha)}, & &\\
N_0 &= \mathcal{B}^{t+1}, & N_1 &= 4\mathcal{B}^{8t+1}, & N &= N_0 N_1,\\
M_c &= N_0 - 1 - (b-1)\ell, & T_c &= M_c + (\mathcal{B}-2)\mathcal{B}^{L_c} N_0, & &\\
J_c &= 2(1+g)^4 D_c + \sum_{i=0}^{8t} \mathcal{B}^{i+1}, & S_c &= g + J_c N_0, & &\\
R_c &= (S_c + T_c + 1)N + T_c + 1, & X &= (N-1)R_c, & Y &= N^2.
\end{aligned}
$$

Every exponent $L_c - e(\alpha)$ is nonnegative. These are literal integer polynomials in f, g, including at assignments outside the useful branch.

Theorem 2 (Source interface)**.** *If P has a nonnegative zero, there are arbitrarily large $f > 0$ with b a square power of two and $b \le g < \mathscr{C}$ such that $Y \mid \binom{2X}{X}$. Conversely, for $f \ne 0$, $b \in \mathrm{Sq}(\mathbb{Z})$ a power of two, and $0 \le g < 2\mathscr{C}$, that divisibility implies a nonnegative zero of P. For all $f \ne 0$, $b \in \mathrm{Sq}(\mathbb{Z})$ and $0 \le g < 2\mathscr{C}$, even before assuming b is a power of two, one has*

$$3 \mid X, \qquad X > 3b, \qquad Y > \max\{b, 256\}.$$

Proof. First establish the bounds for arbitrary g in the indicated interval; it is important not to assume that g already has the desired digit shape. Since $b = 1 + 3f$ is a nonzero square and $f \neq 0$, we have $b \geq 4$. Set $c = 32b$. Then $2b \leq c < \mathcal{B}$, and $1 + g \leq c\mathcal{B}^t$. Lemma 1 and $|c_\alpha| \leq 24H_0$ give

$$0 < D_c \leq 24H_0 \sum_{j=0}^{L_c} \mathcal{B}^j.$$

For the lower inequality, the leading coefficient is $c_0 = 24P(0) \geq 24$, whereas the absolute sum of all lower terms is less than $\mathcal{B}^{L_c}$, because $24H_0 < \mathcal{B} - 1$ and all their exponents are distinct. The scale bound also gives $48H_0c^4 \leq \mathcal{B} - 2$. Consequently

$$\begin{aligned} 0 < J_c &\leq (\mathcal{B}-2)\mathcal{B}^{4t} \sum_{j=0}^{4t} \mathcal{B}^j + \mathcal{B} \sum_{j=0}^{8t} \mathcal{B}^j \\ &< 2\mathcal{B}^{8t+1} < N_1. \end{aligned}$$

Also $0 \leq M_c < N_0$, $g < N_0$, and $0 \leq (\mathcal{B}-2)\mathcal{B}^{L_c} < N_1$. Thus $0 \leq S_c, T_c < N$. As $\beta_S \equiv b \equiv 1 \pmod 3$, $N \equiv 1 \pmod 3$, giving $3 \mid X$. Finally $R_c \geq N + 1$, $X \geq N^2 - 1 > 3b$, and $Y = N^2 > \max\{b, 256\}$.

Suppose now that b is a power of two. Then $\mathcal{B}, N_0, N$ are powers of two. Write $\tau_2(u, v)$ for the number of binary carries in addition. The binary packing identity gives

$$Y \mid \binom{2X}{X} \quad \Longleftrightarrow \quad \tau_2(S_c, T_c) = 0.$$

For completeness, if $N = 2^n$, the identity

$$(N-1)R_c = (S_c + T_c)N^2 + (N - 1 - S_c)N + N - 1 - T_c$$

and binary digit sums show that its digit sum is $2n - \tau_2(S_c, T_c)$. The valuation of $\binom{2X}{X}$ is the digit sum of $X = (N-1)R_c$. It is at least $2n$ exactly when the carry count is zero.

The low N_0 block and the upper block are disjoint, so the same condition is equivalent to

$$\tau_2(g, M_c) = 0, \qquad \tau_2(J_c, (\mathcal{B}-2)\mathcal{B}^{L_c}) = 0.$$

The base-$\mathcal{B}$ digits of M_c are $\mathcal{B} - b$ at the positions w_j and $\mathcal{B} - 1$ elsewhere, through position t. Since $g < 2b\mathcal{B}^t \leq N_0$, the first no-carry condition is exactly

$$g = \sum_{j=1}^{31} n_j \mathcal{B}^{w_j}, \qquad 0 \leq n_j < b.$$

Indeed a binary no-carry addition to $\mathcal{B} - 1$ forces a zero digit, and addition to $\mathcal{B} - b$ forces a digit below b.

For such n, consider

$$H(u) = \left(1 + \sum_j n_j u^{w_j}\right)^4 \sum_\alpha c_\alpha u^{L_c - e(\alpha)} = \sum_{i=0}^{8t} h_i u^i.$$

The coefficient h_{L_c} is $24P(n)$ by Lemma 1 and the multinomial formula. All coefficients satisfy

$$|h_i| \leq 24H_0(1 + \textstyle\sum_j n_j)^4 < \mathcal{B}/2.$$

Here distinct exponent codes ensure that, for a fixed coefficient of the first factor, at most one coefficient of the second factor contributes to a fixed h_i. The number

$$J' = H(\mathcal{B}) + (\mathcal{B}/2) \sum_{i=0}^{8t} \mathcal{B}^i$$

therefore has genuine base-$\mathcal{B}$ digits $\mathcal{B}/2+h_i \in [1,\mathcal{B})$, and $J_c = 2J'$. The second no-carry condition is equivalent, after dividing both inputs by two, to the no-carry addition of J' and $(\mathcal{B}/2-1)\mathcal{B}^{L_c}$. It holds exactly when $\mathcal{B}/2 + h_{L_c} = \mathcal{B}/2$, or $P(n) = 0$. This proves the reverse direction.

For the forward direction, take any zero n. Choose an arbitrarily large even power of two b exceeding all its coordinates; then $b \equiv 1 \pmod 3$ and $f = (b-1)/3 > 0$. The encoded g is less than $b\mathcal{B}^t$. Since $P(0) > 0$, some coordinate is nonzero, so $g \geq \mathcal{B} \geq b$. The preceding equivalences establish the claimed divisibility. □

4 The unchanged ten-integer bridge and finalizer

Theorem 2 supplies precisely the source interface required by Sun's 2026 construction. Its positive bridge, reverse bridge, and full residue interpolation are imported here, not reproved by the finite audit. The following explicit formulas retain the pinned local three-square quintic finalizer unchanged.

The ten unknowns, all unrestricted integers, are (f,g,h,k,l,w,x,y,z,m). All other symbols are polynomial abbreviations:

$$\begin{aligned}
&L = lY, && U = 2LX, && V = 4gwY, && W = bw,\\
&A = U(V+1), && B = 2X+1, && C = B + (A-2)h,\\
&P_n = U^2V, && K = X + 1 + k(P_n - 2),\\
&S = 2A-5, && T = 3WC - 2(W^2-1), && R_2 = (P_n^2-4)K^2 + 4.
\end{aligned}$$

$$\Delta = 8\mathscr{C}^3 gK^2 - g^2\big(32(C-KL)^2\mathscr{C}^3 + g^2K^2\big).$$

$$\begin{aligned}
&O = f^2l^2\Delta, && \xi = zx,\\
&D = (A^2-4)C^2 + 4, && E = C^2D\xi,\\
&F = 4(A^2-4)E^2 + 1, && G = 2(A+2)(A-2)^2E^2 - 1,\\
&H = C + (B-C)F + 16EFy, && I = (G^2-1)H^2 + 1.
\end{aligned}$$

Continue with

$$\begin{aligned}
&r = T + Sy, && \Sigma = \xi S^5, && \Theta = (r^3 - rS^2 - S^3)(r^2 + 23S^2),\\
&a_0 = b\big(1 + (3O-4)b^2z^2\big), && a_1 = R_2, && a_2 = DFI,\\
&v_0 = 1, && v_1 = a_0, && v_2 = a_0(P_n^2 + K^2 + 3),\\
&Z = \Sigma m - \Theta, && A_i = \Sigma^2 v_i^2 a_i \quad (0 \leq i \leq 2),\\
&V_n = Z^2 + A_2 - A_0 - A_1, && W_n = V_n^2 + 4Z^2A_2 - 4A_0A_1.
\end{aligned}$$

The complete equation is

$$\boxed{F_{B_4} = W_n^2 - 16V_n^2Z^2A_2 = 0.} \tag{2}$$

It is division-free and has no variable exponents or unstated parameters.

Theorem 3 (Consistency equation)**.** *Under the inherited calibration* (1) *and the stated Sun bridge theorems,*

$$\mathrm{ZFC} \vdash \big[\neg\exists (f,g,h,k,l,w,x,y,z,m) \in \mathbb{Z}^{10}\ F_{B_4} = 0\big] \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}).$$

Proof. We record why the pinned finalizer applies to the new source rather than silently importing its conclusion for different source polynomials. Sun's positive construction uses only the source interface and produces $b, R_2 \in \mathrm{Sq}(\mathbb{Z})$, $S \mid T$, $C = u_B(A)$, $A > 2$, and $O > 0$. Moreover $O \geq 4$: Δ is a positive multiple of $g \geq b \geq 4$, and f, l are nonzero. Choose a nonzero Pell solution $u^2 - (3O-4)v^2 = 1$ with $b \mid v$, and set $z = v/b$. Such divisibility follows by taking a power of

a Pell unit modulo b; $3O-4>0$ is nonsquare because it is 2 (mod 3). Then a_0 is a square. Full residue interpolation chooses ξ divisible by this z and permits any prescribed residue of y modulo ξ while making DFI square. The polynomial $(q^3-q-1)(q^2+23)$ has a root modulo every nonzero integer, so that residue can ensure $\Sigma \mid \Theta$. Put $x=\xi/z$. All three a_i are then squares, and an integer m makes one factor of the norm in (2) vanish.

Conversely, S is a nonzero odd integer. Neither factor of $(q^3-q-1)(q^2+23)$ has a rational root, so $\Theta \neq 0$. If $\Sigma=0$, the final polynomial is $\Theta^8 \neq 0$; hence any zero has $\xi = zx \neq 0$. Its formal norm identity is

$$F_{B_4} = \prod_{\varepsilon\in\{\pm1\}^3} \left(Z + \Sigma \sum_{i=0}^{2} \varepsilon_i v_i \sqrt{a_i} \right).$$

For the actual a_0 and $z \neq 0$, $|a_0|>1$. Also $\sqrt{|a_1|} \leq P_n^2 + K^2 + 2$, so

$$|v_1| > \sqrt{|a_0|}, \quad |v_2| > \sqrt{|a_0|} + |v_1|\sqrt{|a_1|}.$$

A rational signed sum of these weighted roots forces all roots rational: apply a Galois automorphism changing a root and take the largest changed index. Its term is too large in complex absolute value to be cancelled by earlier terms. Integer radicands with rational roots are nonnegative integer squares. A vanishing factor therefore yields $a_i \in \mathrm{Sq}(\mathbb{Z})$ and $\Sigma \mid \Theta$.

The coprime factors b and $1+(3O-4)b^2z^2$ in a_0 force b to be a positive square and $O>0$. Indeed $b \equiv 1$ (mod 3) excludes negative squares; a zero second factor would force $b=1$, hence $f=O=0$, and then $1-4z^2 \neq 0$. If $O \leq 0$, the second factor is negative. Finally, if a prime has $\nu_p(S) > \nu_p(T)$, the valuations of the two factors of Θ are $3\nu_p(T)$ and $2\nu_p(T)$, contradicting $S^5 \mid \Theta$. Thus $S \mid T$. These are precisely the hypotheses of Sun's reverse bridge: $b, R_2, DFI \in \mathrm{Sq}(\mathbb{Z})$, $O>0$, $\xi \neq 0$, $S \mid T$. It returns a zero of P through Theorem 2. Negating the resulting existence equivalence and using (1) finishes the proof.

The intersective quintic and separating-weight arguments above are retained baseline lemmas. For the quintic, at an odd prime other than 23 either the cubic has a simple root, or its irreducible cubic Frobenius is even and the discriminant -23 is a nonzero square; then the quadratic has a simple root. At 23 the cubic has the simple root 3. At 2, -23 is 1 (mod 8). Hensel lifting and CRT supply roots at every modulus. □

5 Verification, limits, and integration

The top degree of A_2 is $28464t+6766$, strictly exceeding the degrees of Z^2, A_0, A_1. Thus the possible top degree of the final norm is $4 \deg A_2 = 113856t+27064$. Exact leading coefficients on the diagonal specialization modulo 101, 1009, and 1000003 are respectively 95, 909, and 668382. They are nonzero, proving attainment for the fixed tuple.

	Pinned baseline	This source compiler
Integer unknowns	10	10
Equations	1	1
Degree	$126496 \cdot 5^{31} + 27064$	69 554 088 184
Binary additions	566	484
Binary subtractions	25	25
Binary multiplications	851	898
Total binary operations	1442	1407
Fixed leaves	79	79

All fixed powers are expanded into binary gates; reuse and constant leaves are free, as in the baseline. The cost saving is not a claim that fewer multiplications are used. The irregular

exponentiation uses a shared radix-16 fixed-window circuit, and the coefficient polynomial uses homogeneous sparse Horner evaluation.

Run `python3 build_sun_b4.py` followed by `python3 verify_sun_b4.py`. Python's standard library suffices. The checks include exact parent coefficient reconstruction, all payload hashes, all 52,360 four-sums, the 35-term norm identity, exact degree certificates, and 120 modular literal-transcription tests with 5,400 named value comparisons. The last tests are bounded regressions, not proofs of polynomial identity. The circuit construction identities and universal source equivalence are justified by the mathematical proof above.

The imported results are Sun 2026 Theorem 4.1 and Propositions 6.1–6.2, whose source interface is re-established here. Their proofs use Sun 2021 Theorems 4.12–4.13. The number-theoretic proof and endpoint have not been newly formalized in Lean or Rocq. The equation is an explicit candidate with a complete ordinary proof relative to those stated mathematical inputs, not an independent proof of ZFC's consistency.

References

[1] Z.-W. Sun, *Further results on Hilbert's Tenth Problem*, Science China Mathematics 64 (2021), 281–306. https://arxiv.org/html/1704.03504v7.

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[3] ProveIt, immutable commit `22e75dd6b87b1b199bb08c778d7cf075ae5aae08`, `31-sun10/signed10` under the Con(ZFC) report collection. https://github.com/VladimirReshetnikov/ProveIt/tree/22e75dd6b87b1b199bb08c778d7cf075ae5aae08.